

# Rajalakshmi Engineering College

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 8\_Q3

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement

In a user registration system, there is a requirement to implement a username validation module. Users attempting to register must adhere to specific criteria for their usernames to be considered valid.

Your task is to develop a program that takes user input for a desired username and validates it according to the following rules:

The username must not contain any spaces. The username must be at least 5 characters long.

Implement a custom exception, `InvalidUsernameException`, to handle cases where the entered username does not meet the specified criteria.

##### ***Input Format***

The input consists of a string S, representing the desired username.

### **Output Format**

If the username is valid, print "Username is valid: [S]".

If the username is invalid:

1. If the username is short, print "Invalid Username: Username must be at least 5 characters long"
2. If the username contains spaces, print "Invalid Username: Username cannot contain spaces"

Refer to the sample output for formatting specifications.

### **Sample Test Case**

Input: John

Output: Invalid Username: Username must be at least 5 characters long

### **Answer**

```
import java.util.*;
```

```
class InvalidUsernameException extends Exception {  
    public InvalidUsernameException(String message) {  
        super(message);  
    }  
}
```

```
class UsernameValidator {  
    public static void validateUsername(String s) throws  
InvalidUsernameException {  
        if (s.contains(" ")) {  
            throw new InvalidUsernameException("Invalid Username: Username  
cannot contain spaces");  
        } else if (s.length() < 5) {  
            throw new InvalidUsernameException("Invalid Username: Username must  
be at least 5 characters long");  
        }  
    }  
}
```

```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
    String s = sc.nextLine();  
    sc.close();  
    try {  
        validateUsername(s);  
        System.out.println("Username is valid: " + s);  
    } catch (InvalidUsernameException e) {  
        System.out.println(e.getMessage());  
    }  
}
```

**Status :** Correct

**Marks :** 10/10